Computational Pathology and Benchmark Datasets: An Introduction to the Domain and Data

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Abstract

This report introduces the domain of computational pathology and the four public histopathology datasets it covers. It first explains what diagnostic pathology is, how a stained glass slide becomes a gigapixel whole-slide image, why such images are tiled into patches, and how weakly supervised and frozen foundation-model pipelines turn those patches into predictions. It then describes, in detail, four H&E datasets — CAMELYON16, TCGA-UT, BRACS, and PANDA — presented in the order of the diagnostic question they pose, from detection (*is cancer present?*) through cancer typing and lesion subtyping to ordinal grading (*how aggressive is it?*). Each dataset is given with its clinical background, an example patch, a summary table, and its class distribution.

Contents

1 Computational Pathology

Diagnostic pathology is the branch of medicine that examines tissue under the microscope to detect and characterise disease. When cancer is suspected, a biopsy or surgical resection is fixed, thinly sectioned, mounted on a glass slide, and stained — most commonly with hematoxylin and eosin (H&E), which colours cell nuclei blue-purple and cytoplasm and stroma shades of pink — so that a pathologist can read cellular and tissue morphology and render a diagnosis. This reading is the clinical reference standard for cancer detection, typing, and grading, and it drives treatment decisions. It is also labour-intensive, requires scarce expertise, and is subject to inter- and intra-observer variability, which motivates computational assistance (Madabhushi and Lee, 2016; Song et al., 2023).

Computerised analysis of microscopy images is not new: the earliest work on cell and chromosome image analysis dates back more than fifty years. What changed recently is scale and method. *Digital pathology* digitises the glass slide with a whole-slide scanner, enabling telepathology, archiving, and education; *computational pathology* (CPath) goes further, treating the resulting images as high-dimensional data for quantitative, reproducible analysis and integrating them with clinical and molecular context. A useful distinction here is between the *ground truth*, the abstract label one wishes to predict, and the *gold standard*, the practical benchmark used to approximate it — typically a pathologist's own assessment. Because algorithmic outputs can eventually prove more reproducible than the human benchmark that trained them, CPath must navigate a *gold-standard paradox* in which the reference itself is imperfect (Abels et al., 2019; Madabhushi and Lee, 2016; Hosseini et al., 2024).

The clinical value of CPath is usually framed along two axes. *Automation* standardises tasks pathologists already perform — detecting metastases, subtyping and grading tumours, quantifying immunohistochemical markers — reducing workload and observer variability. *Discovery* identifies signatures beyond human visual limits, such as predicting prognosis, therapy response, or molecular alterations (for example microsatellite instability) directly from H&E morphology; these are the "sub-visual" features that quantitative analysis can mine but a human eye cannot reliably see (Song et al., 2023; Madabhushi and Lee, 2016). Table 1 summarises representative tasks. The datasets in this report all sit on the automation axis: each carries a diagnostic label a pathologist established, and the task is to predict it from H&E tissue.

Role	Representative tasks
Automation	Metastasis detection; cancer-type classification; tumour subtyping; Gleason/ISUP grading; mitosis counting; immunohistochemistry (PD-L1, Ki-67) quantification
Discovery	Prognosis and recurrence-risk prediction; molecular-marker prediction (e.g. microsatellite instability) from H&E; therapy-response biomarkers

Table 1: Two roles of computational pathology, following Song et al. (2023) and Madabhushi and Lee (2016). The datasets in this report are automation tasks with pathologist-established labels.

1.1 The whole-slide imaging pipeline

A digitised slide is a *whole-slide image* (WSI): a single H&E image of an entire tissue section, routinely on the order of 100,000×100,000 pixels and 0.5–4 GB in size. To make this tractable, a WSI is stored as a *pyramid* of progressively downsampled copies, so the same slide can be read at several magnifications (Table 2): high magnification resolves nuclear detail and mitoses, while lower magnifications carry the tissue-architectural context that diagnosis also depends on. Scanning the whole section, rather than a manually chosen field of view, removes selection bias and captures intratumoral heterogeneity, but it produces far more data than fits in GPU memory (Song et al., 2023; Abels et al., 2019).

Magnification	Resolution	What it resolves
40×	≈0.25 μm/pixel	Nuclear detail, chromatin, mitotic figures
20×	≈0.5 μm/pixel	Cellular morphology, individual glands
10×	≈1.0 μm/pixel	Local tissue architecture
5×	≈2.0 μm/pixel	Macro-architectural context

Table 2: Typical magnification levels of a pyramidal WSI and the morphology each makes legible (Song et al., 2023). High magnifications are needed for nuclear grading, lower ones for architectural context.

The standard response to the size of a WSI is a divide-and-conquer strategy: the image is partitioned into small *patches* (also called *tiles*), for example 256 × 256 pixels, each processed independently and later aggregated into a slide- or patient-level prediction (Song et al., 2023). Figure 1 sketches the resulting pipeline, from stained slide to prediction, that every dataset in this report passes through.

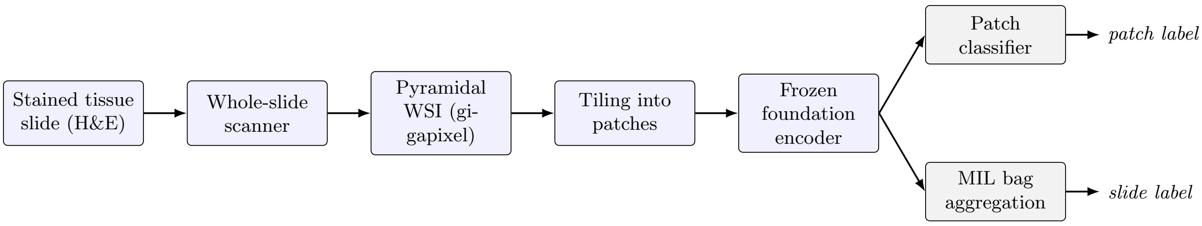


Figure 1: The computational-pathology pipeline the datasets pass through. A stained slide is scanned into a pyramidal WSI and tiled into patches; a frozen foundation-model encoder embeds each patch once, after which either a patch classifier predicts a per-tile label or a multiple-instance head aggregates the patch features of a slide into a slide-level prediction.

1.2 Learning from patches: weak supervision and foundation models

How labels attach to these units determines the learning regime. When each patch carries a reliable label — because it was cropped from a pathologist-selected region, or intersected with an expert pixel mask — ordinary patch classification applies. When only a slide-level diagnosis is available, the problem is *weakly supervised*: the slide is a *bag* of patch *instances* that share one label, and *multiple-instance learning* (MIL) aggregates the instances — typically through a learned attention mechanism that weights diagnostically salient patches — into a slide prediction (Song et al., 2023; Hosseini et al., 2024). Because tiling discards spatial context, context-aware models such as graph neural networks (over patches or nuclei) and transformers (with global self-attention) are sometimes used to recover it, but the attention-MIL bag remains the workhorse for slide-level tasks. Both regimes appear in the datasets below, and two of them (CAMELYON16 and PANDA) support both on the same slides.

The features fed to these heads have shifted from hand-crafted descriptors — interpretable measurements of nuclear shape, texture, or spatial architecture — to learned representations that capture patterns beyond human perception. Modern CPath rarely trains an image encoder from scratch: a pathology *foundation model*, pre-trained self-supervised on large slide archives, provides a frozen feature extractor that embeds each patch once into a fixed-length vector, leaving only a lightweight classifier or MIL head to train. Virchow2 is one such H&E foundation model (Vorontsov et al., 2024; Zimmermann et al., 2024). Freezing the representation lets a study isolate the effect of a downstream modelling choice from the confounds of end-to-end backbone training, and makes the patch and MIL regimes directly comparable, because both consume the same embeddings.

1.3 Data and annotation

Deep learning is data-hungry, and expert annotation is the binding constraint in CPath: pathologist labelling is slow, expensive, and hard to scale, so datasets range from densely mask-annotated cohorts to those carrying only a single slide-level diagnosis (Abels et al., 2019; Hosseini et al., 2024). The four datasets in this report deliberately span that range of annotation form, from pixel-level tumour masks through region-of-interest crops to a single slide-level diagnosis. The next section introduces them in the order of the diagnostic question they pose.

2 Datasets

Four public H&E datasets are used, described here in the order of the diagnostic question they pose — an arc that also escalates the structure of the label and varies the organ and the form of annotation:

- **CAMELYON16** — *is cancer present?* Binary metastasis *detection* in sentinel lymph nodes, with pixel-mask annotations.
- **TCGA-UT** — *what cancer is it?* Pan-cancer *typing* across 32 organs, each patch labelled with its slide's cancer type.
- **BRACS** — *how abnormal is this tissue?* Breast-lesion *subtyping* along the benign–atypical–malignant continuum, with region-of-interest annotations.
- **PANDA** — *how aggressive is it?* Ordinal ISUP *grading* of prostate biopsies, with region masks and built-in two-provider heterogeneity.

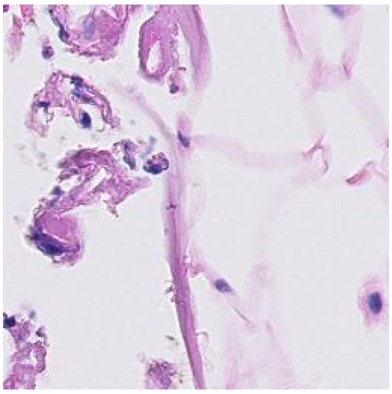
2.1 CAMELYON16 — lymph-node metastasis detection

CAMELYON16 is an organ-specific H&E cohort of sentinel lymph-node sections for the detection of breast-cancer metastases (Ehteshami Bejnordi et al., 2017; Litjens et al., 2018). The diagnostic question is the most basic in oncology — *is cancer present?* — and the label is binary: each slide is either *normal* or contains *tumor* (metastasis). Expert annotations are supplied as pixel-level masks marking tumour tissue, rather than as cropped regions, so individual tiles can be labelled by intersecting them with the mask.

The benchmark uses 398 of the 399 published slides (160 tumor, 238 normal) after excluding one slide without usable tile evidence; each slide is its own patient, so slide-disjoint splits coincide with patient-disjoint splits. Because metastatic tissue occupies only a small fraction of a sectioned node, mask-derived tumor tiles are scarce: the patch distribution is 2,340,040 normal against 59,672 tumor tiles (Table 3, Figure 2), even though tumour and normal slides are themselves close to even (160 versus 238)—tumour tiles are scarce even on slides that do contain a metastasis. The 20 tumor slides whose annotations are not exhaustive are excluded from the tile-labelling regime, where unannotated tumour could be mislabelled as normal, but retained where only the reliable slide-level label is used.

2.2 TCGA-UT — pan-cancer typing

TCGA-UT widens the lens from one organ to the whole body. It is derived from diagnostic WSIs in The Cancer Genome Atlas (TCGA), a pan-cancer resource linking molecular and clinical data with large image collections (Tomczak et al., 2015). The TCGA-UT release turns this slide archive into a computational-pathology dataset by cropping H&E tumour patches from pathologist-selected uniform tumour regions: 1,608,060 RGB patches of 256 × 256 pixels from 8,736 diagnostic slides, across 32 cancer types (Komura et al., 2022). Every patch carries



Property	Value
WSIs (tumor / normal)	160 / 238
Patch-labelled WSIs	378
WSI-bag size (median tiles)	7,874
Tile labels	tumor, normal
Tumor / normal tiles	59,672 / 2,340,040

Table 3: CAMELYON16 dataset properties, alongside an example 256 × 256 tumor tile. The WSI-bag size is the data-driven median of available tissue tiles per slide. Tumour tiles are scarce even within slides that contain a metastasis.

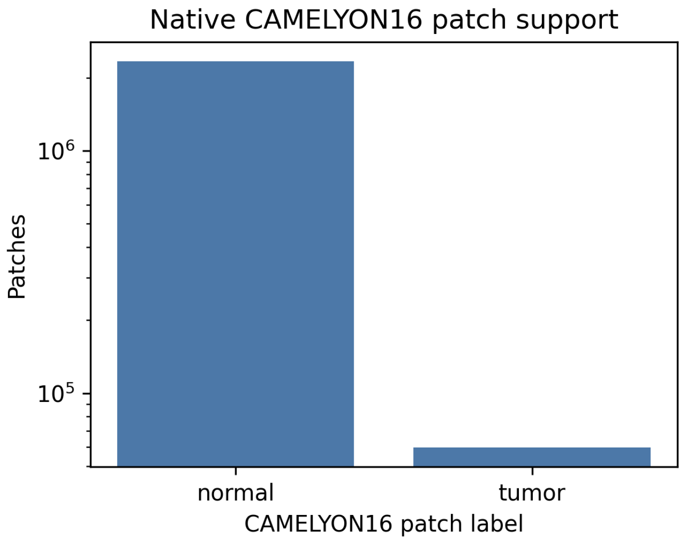


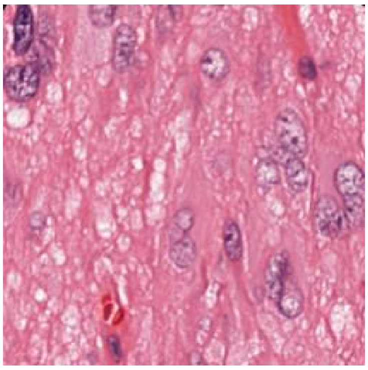
Figure 2: CAMELYON16 patch support by tumor/normal label after mask-based labelling (log scale). Tumor tiles are far scarcer than normal tiles.

the cancer-type label of the slide it was cropped from, so the patches are themselves labelled samples — the unit of patch-level classification — while their slide identities also allow grouping into slide-level bags. The labels are cancer types, not dense pixel masks or cell-level annotations, so the diagnostic question is *what cancer is it?*, a nominal classification across 32 organs.

The largest class is glioblastoma multiforme (GBM; 792 slides) and the smallest is diffuse large B-cell lymphoma (DLBC; 28 slides), with the remaining types spread across a gradually declining tail (Table 4, Figure 3).

2.3 BRACS — breast-lesion subtyping

BRACS narrows back to a single organ but sharpens the question from *which cancer* to *how abnormal is this tissue*. It is an H&E breast-pathology dataset of 547 WSIs and 4,539 regions of interest (ROIs) from 189 patients, annotated by consensus of three board-certified pathologists (Brancati et al., 2022). The labels form three lesion types — benign, atypical, and malignant — refined into seven subtypes: normal (N), pathological benign (PB), usual ductal hyperplasia (UDH), flat epithelial atypia (FEA), atypical ductal hyperplasia (ADH), ductal carcinoma in



Property	Value
Cancer-type labels	32
Diagnostic slides	8,736
Patches	1,608,060 of 256 × 256
Largest class	Glioblastoma (GBM), 792 slides
Smallest class	B-cell lymphoma (DLBC), 28 slides

Table 4: TCGA-UT dataset properties, alongside an example 256 × 256 H&E tumor patch. Because every patch inherits its slide's cancer-type label, the patch- and slide-level class distributions coincide.

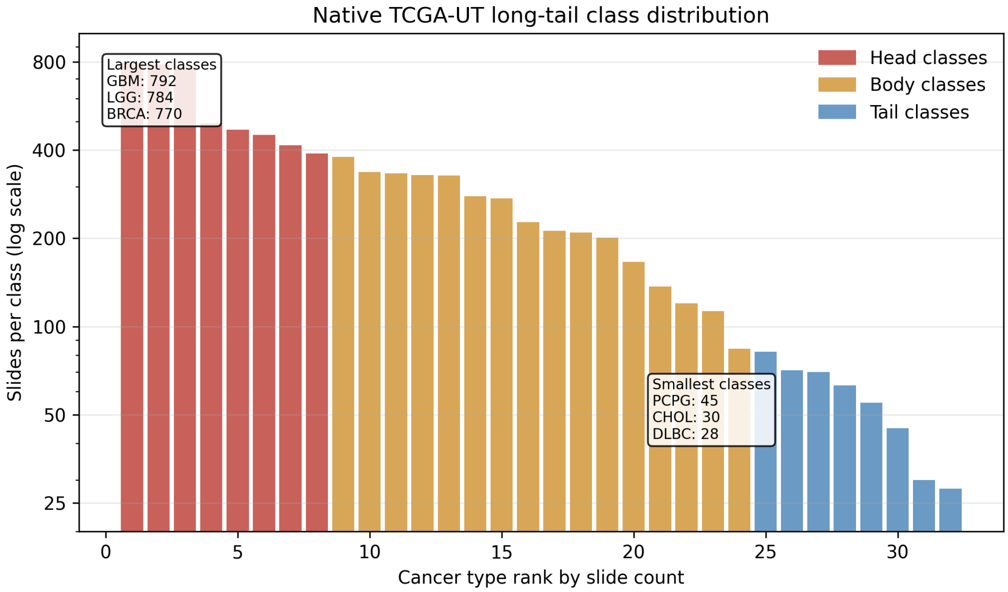


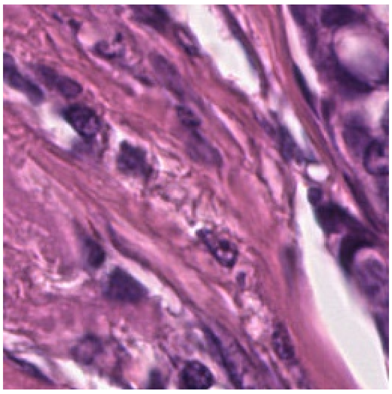
Figure 3: TCGA-UT slide support by cancer-type rank, declining gradually across the 32 classes from GBM at the head to DLBC at the tail.

situ (DCIS), and invasive carcinoma (IC). Annotations are ROI crops rather than whole-slide labels or pixel masks, and a single slide may contain ROIs of several subtypes.

The benchmark uses the seven-subtype target, which keeps clinically important but comparatively scarce precancers visible: the atypical lesions FEA and ADH would be hidden by collapsing to three lesion types. After deterministic ROI tiling into 256 × 256 patches, 387 of the 547 WSIs contribute at least one annotated tile; the prepared cohort spans 151 patients, the full 4,539 ROI records, and 99,352 patches. Counting the slides that contain each subtype (a slide with several lesions counts under each), support ranges from PB (196 slides) at the head to DCIS (87 slides) at the tail (Table 5, Figure 4). BRACS's difficulty comes from fine-grained morphology rather than a steep tail.

2.4 PANDA — prostate cancer grading

PANDA (Prostate cANcer graDe Assessment) poses the most nuanced question — *how aggressive is the cancer?* — as an ordinal grading problem. It is an H&E dataset of prostate core-needle biopsies assembled for automated Gleason grading and the largest public cohort



Property	Value
Patients	151
WSIs with ROI tiles	387
ROI tiles	99,352
WSI-bag size (median tiles)	360
Subtype labels	N, PB, UDH, FEA, ADH, DCIS, IC

Table 5: BRACS dataset properties, alongside an example 256 × 256 ROI tile. Slide-level support counts each slide under every subtype it contains; the tile-level distribution instead weights subtypes by their tile counts.

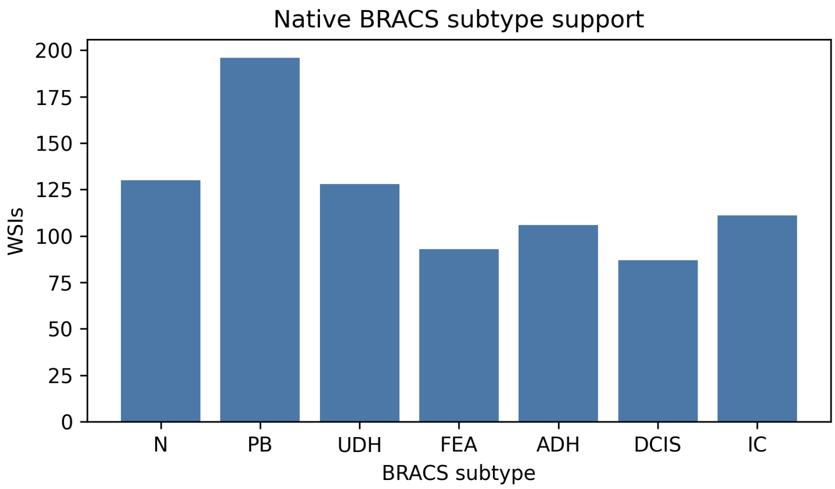


Figure 4: BRACS WSI support by seven-subtype label (slides containing each subtype), from PB at the head to DCIS at the tail.

of its kind, with 10,616 whole-slide biopsies (Bulten et al., 2022). Gleason grading describes prostate-cancer severity from glandular growth patterns; pathologists summarise the dominant and secondary malignant patterns into ordinal ISUP Grade Groups, where higher groups indicate more aggressive morphology. Here grade 0 marks benign tissue and grades 1–5 are the ordered ISUP cancer groups.

PANDA differs from the other three along every axis a generality check exercises: a third organ (prostate), an ordinal rather than nominal or binary label, and a built-in source of heterogeneity, because the slides come from two institutions — Radboud University Medical Center and Karolinska Institute — that differ in staining, scanning, and annotation convention, and grade differently (Karolinska biopsies concentrate at low grades, Radboud spread more evenly). To keep frozen-feature extraction tractable on a cohort an order of magnitude larger than the others, the benchmark draws a stratified subset of approximately 2,000 biopsies preserving the joint distribution of ISUP grade and provider. The ISUP distribution runs from ISUP 0 (544 biopsies) at the head to ISUP 5 (230) at the tail. PANDA's masks additionally support a provider-consistent binary cancer/benign tile label — the only labelling the two institutions share — giving 580,531 benign against 122,751 cancer tiles, so the same biopsies furnish both an ordinal slide-level grading target and a binary tile-level detection target (Table 6, Figure 5).

PANDA is the only one of the four datasets whose two regimes answer different questions, and this follows from its annotations rather than from a design choice. It is the one cohort carrying two independent label sources at two granularities: a slide-level ordinal ISUP grade, which summarises the whole biopsy and cannot be assigned to a single tile, and pixel masks that yield a per-tile label. CAMELYON16, TCGA-UT, and BRACS instead each provide a single label concept — tumour/normal, cancer type, or lesion subtype — that applies unchanged at both tile and slide level, so their two regimes share one task and differ only in prediction unit.

Property	Value
WSIs (biopsies)	2,000
Patch-labelled WSIs	1,984
WSI-bag size (median tiles)	421
WSI labels / tile labels	ISUP 0–5 / benign, cancer
Benign / cancer tiles	580,531 / 122,751

Table 6: PANDA dataset properties after stratified subset selection and mask-based patch labelling. The WSI-bag size is the data-driven median of available tissue tiles per biopsy. Unlike the other datasets, the tile and slide regimes carry different labels (binary cancer/benign vs six-class ISUP), and so describe two different tasks.

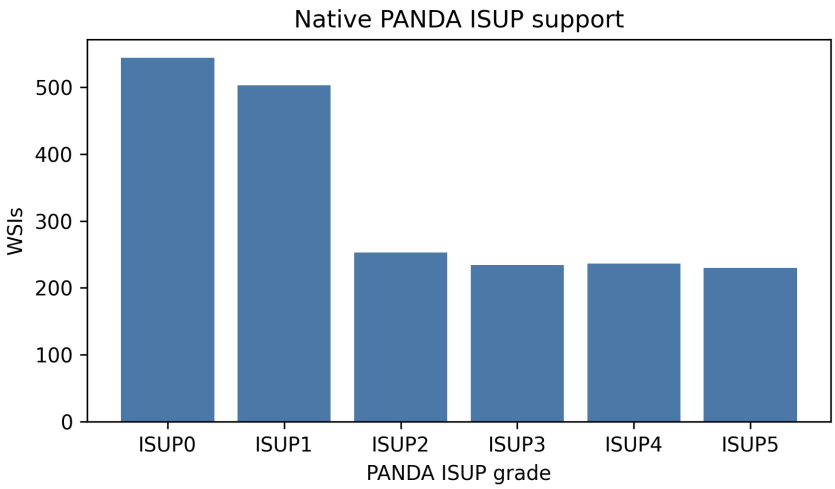


Figure 5: PANDA WSI support by six-class ISUP grade after stratified subset selection, from ISUP 0 at the head to ISUP 5 at the tail.

3 Summary

This report introduced the computational-pathology pipeline — from a stained glass slide to a gigapixel whole-slide image, patching, weakly supervised multiple-instance learning, and frozen foundation-model features — and the four H&E datasets it covers, presented as a progression of diagnostic questions from CAMELYON16 detection through TCGA-UT typing and BRACS subtyping to PANDA grading, each with its organ, label type, annotation form, example patch, and class distribution.

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